

Hardware Safety Requirements and Design

(System Component: „Power Unit and Motor Monitoring“)

1 Introduction

This specification refines the allocated component safety requirements down to hardware level. Requirements should be refined as much as it is necessary to derive a hardware design (hardware circuit).

The following tables only contain the hardware safety requirements. In reality of course many additional non requirements to the hardware have to be described.

2 Hardware Safety Requirements

If necessary additional HW safety requirements are to be defined after the design has been verified by mean of probabilistic calculations (HW-metrics)

Reference from TSC	ID	Description	ASIL
COMPR1	HWSR_01	The motor current shall be measured continuously by using a shunt resistor	B
COMPR2	HWSR_02	The measured motor current shall be continuously transmitted to the motor ECU as standardized signal (0,1 – 10V)	B
COMPR3	HWSR_03	Power shall be switched off by a power relay	B
COMPR3	HWSR_04	The power relay shall be able to cut the energy supply to the motor within 0,5s	B
COMPR4	HWSR_05	Switch off shall be defined by 0V	B
COMPR4	HWSR_06	Switch on shall be defined by 5V	B

3 Hardware Design

If necessary additional HW safety requirements are to be defined after the design has been verified by means of probabilistic calculations (HW metrics calculation)

